About Simulation-to-Reality Correlation Standard - (SRCS)

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Architecture: SIMULOS® — Simulation Governance Architecture Category: Governance & Enforcement Subcategory: Simulation Governance Parent Standard: Simulation-to-Reality Correlation Standard (SRCS) Governed Space: Simulation-to-Reality Correlation Governance Flow Position: 10 of 12 — Simulation Output Governance → Simulation-to-Reality Correlation → Downstream Simulation Governance

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Canonical Definition

Simulation-to-Reality Correlation Standard (SRCS) defines the governance conditions under which simulated behavior, conditions, events, relationships, and outputs are systematically compared with corresponding Operational Reality to determine where meaningful correlation exists, where material deviation occurs, how reality alignment should be characterized, and how uncertainty and limitations constrain any claim that simulation represents reality.

SRCS does not assume that a technically sophisticated simulation accurately represents the real world. It establishes a governed relationship between what occurred in simulation and what is observed, measured, or otherwise established in Operational Reality.

Why SRCS Matters

Organizations increasingly rely on simulation to examine systems before, during, and sometimes instead of direct real-world testing.

The critical governance question therefore becomes not merely:

Did the simulation run correctly?

but:

How closely does what happened in simulation correspond to what actually happens in reality?

A simulation may be internally consistent, computationally precise, repeatedly executable, and capable of producing enormous volumes of output while still diverging materially from reality.

SRCS makes that divergence visible and governable.

Core Principle

Simulation similarity is not reality equivalence.

A simulation may correlate strongly with reality for one: behavior, operating condition, environment, system state, scenario, time period, or output, while correlating poorly in another.

Correlation must therefore remain bounded by what was actually compared.

Governance Flow

Simulation Outputs

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Simulation-Reality Comparison

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Correlation Criteria

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Simulation-Reality Deviation

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Reality Alignment Assessment

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Correlation Uncertainty

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Bounded Simulation-to-Reality Correlation Record What SRCS Governs

SRCS governs the relationship between simulated and real-world observations through five distinct questions:

SRCM — Simulation-Reality Comparison Module What simulated and real-world phenomena can legitimately be compared?

SCCM — Simulation Correlation Criteria Module Against which criteria should correspondence between simulation and reality be assessed?

SDMM — Simulation-Reality Deviation Module Where and by how much does simulation materially diverge from reality?

SRAM — Reality Alignment Module To what extent does the simulation remain aligned with the relevant Operational Reality?

SCUM — Correlation Uncertainty Module How much uncertainty remains in the claimed simulation-to-reality relationship?

Bounded Correlation

SRCS prevents a narrow correlation result from becoming a universal representation claim.

For example:

Strong correlation in nominal operating conditions

does not establish:

Strong correlation under extreme conditions, failures, rare events, different environments, or future configurations.

Every material correlation claim should therefore remain connected to its:

Object + Reality Reference + Conditions + Criteria + Observations + Deviations + Uncertainty + Scope Relationship to ORA™

ORA™ governs Operational Reality.

SIMULOS® does not replace that function.

SRCS governs the correlation relationship between the simulation and the relevant Operational Reality supplied, observed, or described through the appropriate reality-governance basis.

The architectural distinction is:

ORA™ → What is occurring in Operational Reality?

SIMULOS® SRCS → How does the simulation correspond to that reality?

Relationship to VALIDOS®

SRCS does not itself declare a simulation universally valid.

It produces governed correlation evidence that may later contribute to validation under VALIDOS®.

Therefore:

Simulation-to-Reality Correlation ≠ Validation Determination.

A high correlation in one governed space may be important validation evidence without establishing universal validity.

Relationship to INTEGROS®

SRCS does not take ownership of general integrity governance.

INTEGROS® may govern integrity properties of evidence, records, measurements, or other artifacts used in the correlation process.

SRCS governs what those simulation and reality observations reveal about their relationship. Use Case 1 — Space Mission Simulation

A space agency develops a simulation environment for spacecraft navigation, autonomous recovery, communication loss, propulsion behavior, and subsystem failures.

After real mission or hardware-test data becomes available, SRCS can compare:

Simulated Behavior ↔ Observed Reality

The analysis may reveal strong alignment for nominal navigation but significant deviation during combined sensor degradation and communication delay.

The important result is therefore not simply:

“The simulation correlates with reality.”

It is:

where it correlates, where it deviates, under which conditions, and with what remaining uncertainty.

That information can guide subsequent model improvement, scenario design, simulation reliance, and validation. Use Case 2 — Autonomous System Deployment

An autonomous vehicle or industrial robot performs successfully across millions of simulated scenarios before deployment.

Operational deployment then produces real-world observations involving: sensor noise, unexpected human behavior, environmental variability, timing differences, physical wear, rare interactions, and conditions insufficiently represented in simulation.

SRCS compares the simulated and operational behavior and identifies where the simulation-to-reality gap becomes material.

This allows the organization to distinguish:

high simulation performance

from:

demonstrated correspondence with Operational Reality.

Enterprise Value

SRCS can operate over an organization's existing simulation and operational infrastructure.

It does not require replacement of the simulator.

It provides a governance framework for answering questions such as:

Which simulated behaviors have actually been compared with reality?

Against what correlation criteria?

Where are the largest deviations?

Under which operating conditions does alignment deteriorate?

Which parts of the simulation remain uncorrelated?

How uncertain is the correlation claim?

Can the organization legitimately rely on correlation established in one environment when operating in another?

This turns the frequently discussed simulation-to-reality gap into a structured governable space.

Foundational Principle

Simulation similarity is not reality equivalence. Correlation becomes governable only when correspondence, deviation, alignment, uncertainty, and the boundaries of the comparison remain explicitly visible.

Purpose of SRCS

Simulation-to-Reality Correlation Standard (SRCS) provides the governance layer through which organizations can determine not whether a simulation merely looks realistic, but where, under what conditions, to what extent, and with what uncertainty its behavior corresponds to Operational Reality.

It establishes the bridge between simulated reality and observed reality without confusing correlation with equivalence, validation, or universal truth.

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Canonical Source

https://originopenfoundation.org/